COMP90007 Internet Technology

Week4

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Sampling

- **Sampling** is a fundamental bridge between continuous signals and discrete (digital) signals. It helps us to transform continuous signals into digital values.
- Nyquist theorem (noiseless channel)
  - Maximum data rate = $2B \times \log_2 V \ \ bits/sec$
  - **If a signal has bandwidth B, then the signal can be fully reconstructed by sampling with at least 2B rate.**
- Shannon theorem ("noise-related")

Nyquist's theorem (**Noiseless**)

- Nyquist proved that if an arbitrary signal has been run through a low-pass filter of bandwidth *B*, the filtered signal can be completely reconstructed by making only 2*B* (exact) samples per second.
- Sampling the line faster than 2*B* times per second is pointless because the higher-frequency components that such sampling could recover have already been filtered out.
- If the signal consists of *V* discrete levels, Nyquist's theorem states:

Maximum data rate = $2B \times \log_2 V \ \ bits/sec$

Shannon's theorem (**noise-related**)

- This theorem relates the data rate to the bandwidth (B) and signal strength (S) relative to the noise (N):

Maximum data rate = $B \times \log_2(1 + S/N)\ bits/sec$

How fast signal can change

How many levels can be seen

Sampling

- **Frequency**: the number of oscillations per second of a wave is called frequency, measured in Hertz
- The **second concept about bandwidth**: the frequency range in the given medium – continuous signal

Question 1 (Sampling)

- Consider a **telephone signal** that is bandwidth limited to 4 kHz.
  - (a) At what rate should you sample the signal so that you can completely reconstruct the signal?
  - (b) If each sample of the signal is to be encoded at 256 levels, how many **bits/symbol** are required for each sample?
  - (c) What is the **minimum** bit rate required to transmit this signal?

Solution 1 (Sampling)

- (a) At what rate should you sample the signal so that you can completely reconstruct the signal?

  By Nyquist's Theorem:

  min. sampling rate = $2 \times 4000 = 8$ kHz = 8000 samples/s

Solution 1 (Sampling)

- (b) If each sample of the signal is to be encoded at 256 levels, how many **bits/symbol** are required for each sample?

256 possible values per sample requires $\log_2 256 = 8\ bits/sample$

Solution 1 (Sampling)

- (c) What is the **minimum** bit rate required to transmit this signal?

8 bits/sample × 8000 samples/s = 64 kbps

Question 2 (Sampling)

- Is the Nyquist theorem true for optical fibre or only for copper wire?

Solution 2 (Sampling)

- The Nyquist theorem is a property of mathematics and has nothing to do with technology.
- It states that if you have a function whose Fourier spectrum (frequency domain representation) **does not contain** any frequency components (sines or cosines) **above *f*, then by sampling at a frequency of *2f*,** you capture all the information there is. The Nyquist theorem is **independent of the transmission medium.**

Question 3 (Sampling)

- A **noiseless 4 kHz channel** is sampled every 1 ms. What is the maximum data rate of the communications channel?

Solution 3 (Sampling)

- A noiseless channel can carry an arbitrarily large amount of information, no matter how often it is sampled. (i.e. there can be an infinite number of signalling levels because there is no noise)

Solution 3 (Sampling)

- Just send a lot of data per sample. Assume a **4 kHz channel, sampled at 8 kHz.** If each sample is **16 bits**, the channel can send **128 kbps**. If each sample is 1024 bits, the channel can send 8.2 Mbps.
- *(16/T -> 16*f -> 16bits * 8kHz -> 128kbps)*
- The key word here is “noiseless.” With a normal noisy 4 kHz channel, Shannon specifies a limit on the information rate on the channel known as its capacity.

Question 4a (Sampling)

- The **bandwidth** of a television video stream is 6 MHz. How many bits/sec are sent if **four-level digital signals are used**? Assume a noiseless channel.

Solution 4a (Sampling)

- Using the Nyquist theorem, we should sample at 12 MHz or 12 million samples/s.
- Four levels of signalling provide:
  - $\log_2 4 = 2\ bits/sample\ (V = 4)$
- Hence, the total data rate is:
  - $12\ million\ samples/s \times 2\ bits/sample = 24\ Mbps$

Question 4b (Sampling)

- The bandwidth of a television video stream is 6 MHz. How many bits/sec are sent if four-level digital signals are used? Now assume a S/N of 20db.

Solution 4b (Sampling)

- Using the Nyquist theorem, we should sample at 12 MHz or 12 million samples/s.
- Using Shannon's theorem, we have:

$$B \times \log(1 + S/N)$$

Solution 4b (Sampling)

- $B \times \log\left(1 + \frac{S}{N}\right) = 6\ MHz \times \log(1 + 100)$
- $= 6\ MHz \times 6.65$
- $= 39.9$ Mbits/sec

Question 5

- The following character encoding is used in a data link protocol:

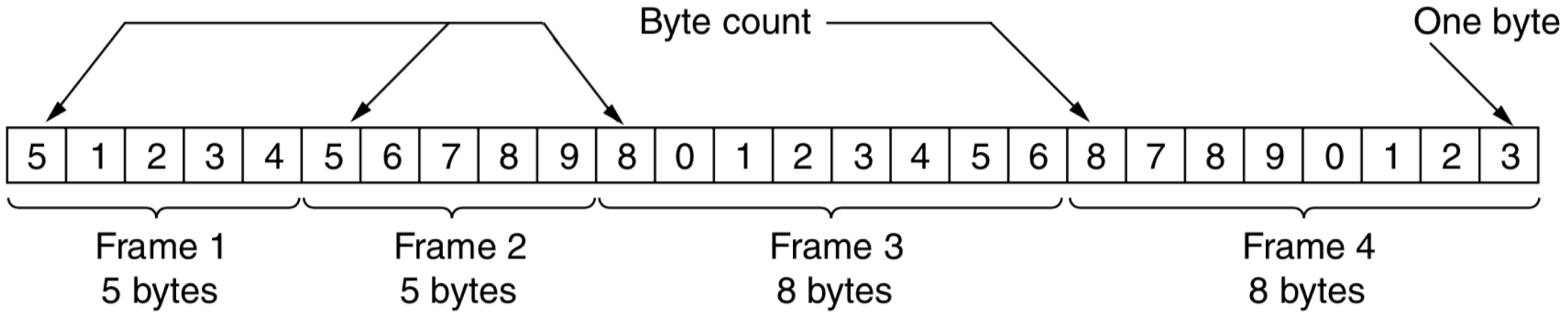
A:	`01000111`	B:	`11100011`
FLAG:	`01111110`	ESC:	`11100000`

- Show the bit sequence transmitted (in binary) for the four-character frame payload *A B ESC FLAG,* when each of the following framing methods are used:
  (a) Character count
  (b) Flag bytes with byte stuffing
  (c) Starting and ending flag bytes, with bit stuffing

Character count

- Use a field in the header to specify the number of bytes in the frame.
- When the data link layer at the destination sees the byte count, it knows how many bytes follow and hence where the end of the frame is.

Character count



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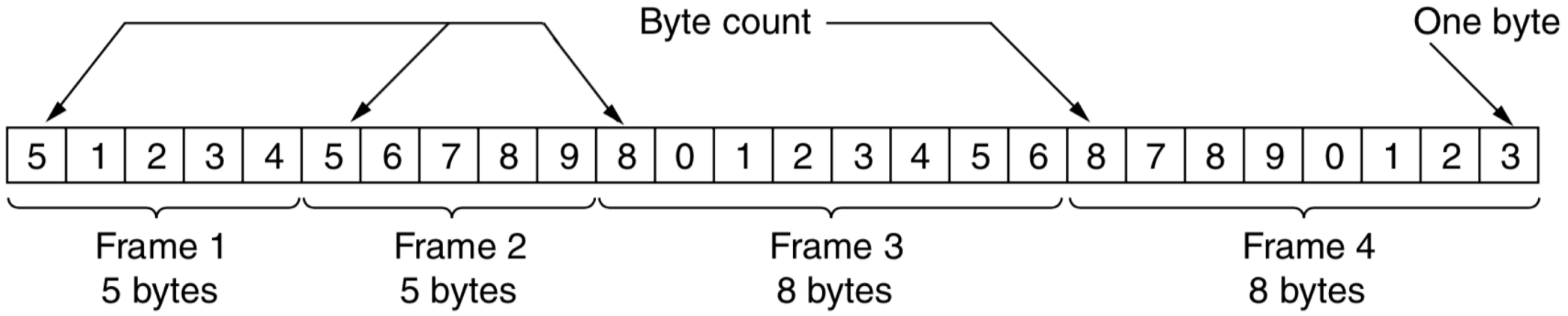
Character count

- Example:
  - The byte count of 5 in the second frame becomes a 7 due to a single bit flip.



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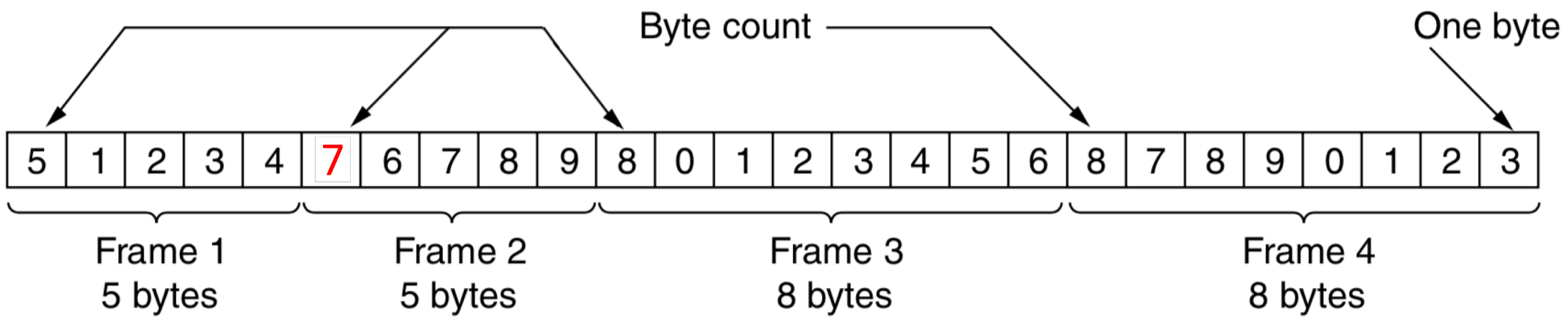

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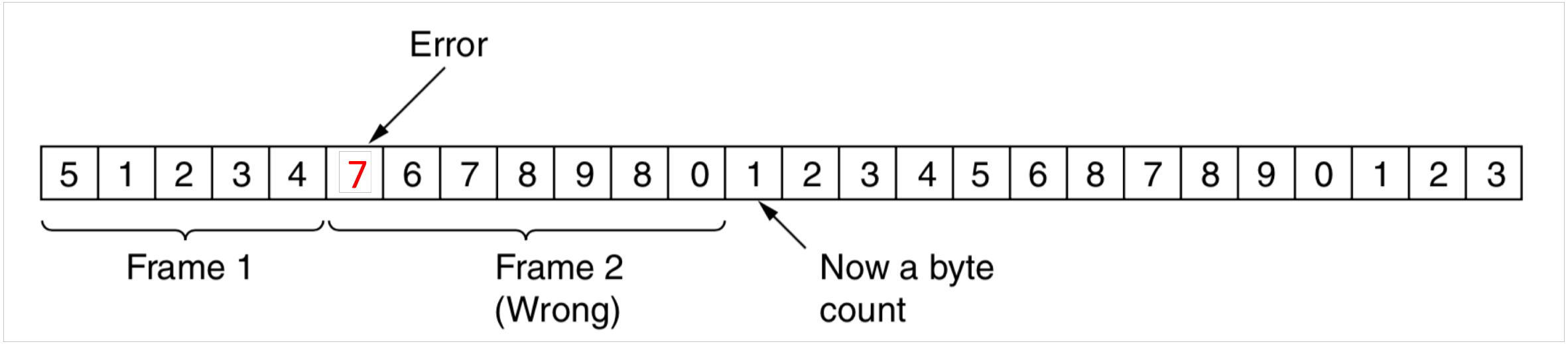
Character count

- Example:
  - It will then be unable to locate the correct start of the next frame.



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Flag Byte

- The second framing method gets around the problem of resynchronization after an error by having each frame start and end with special bytes. Often the same byte, called a **flag byte**, is used as both the starting and ending delimiter.

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Flag Byte

- **Two consecutive flag** bytes indicate the end of one frame and the start of the next. Thus, if the receiver ever loses synchronization it can just search for **two flag bytes** to find the end of the current frame and the start of the next frame.

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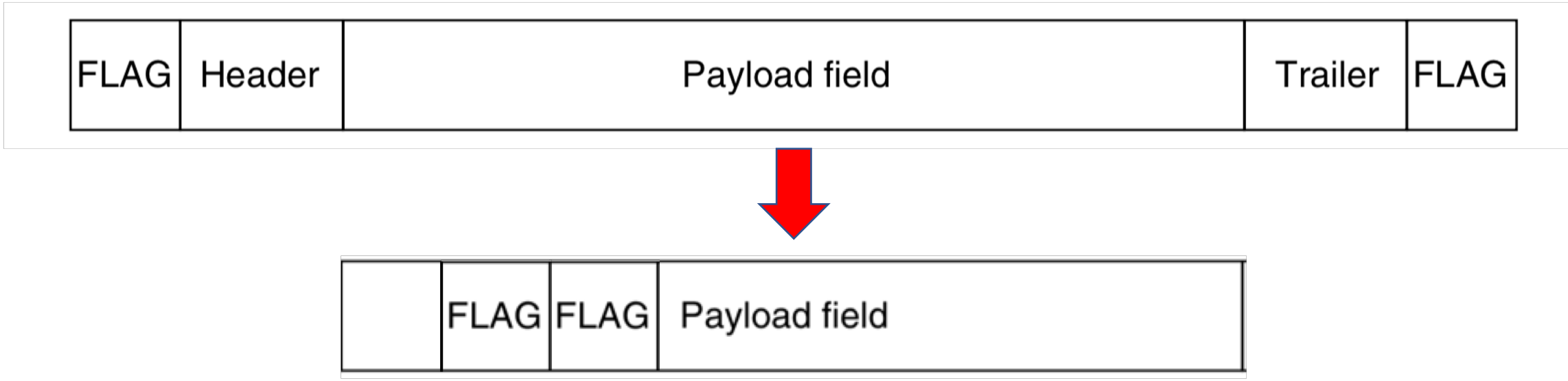
Flag Byte

- Drawbacks: The flag byte occurs in the data



Data

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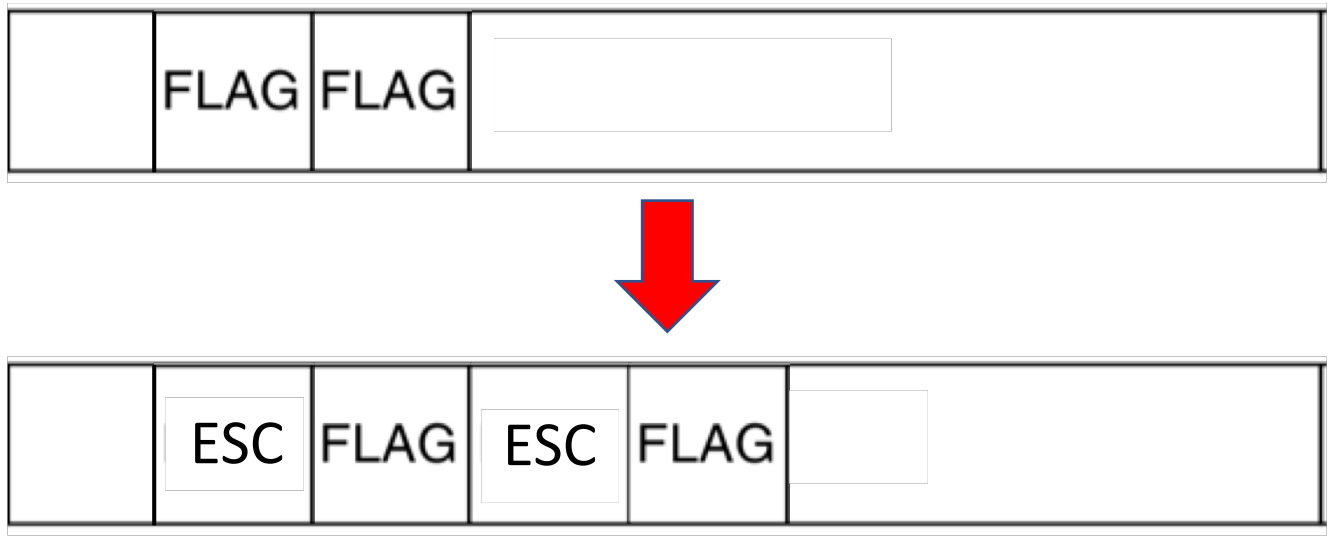
Byte Stuffing

- Solution: Insert a special escape byte (ESC) just before each ‘‘accidental’’ flag byte in the data.



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Byte Stuffing

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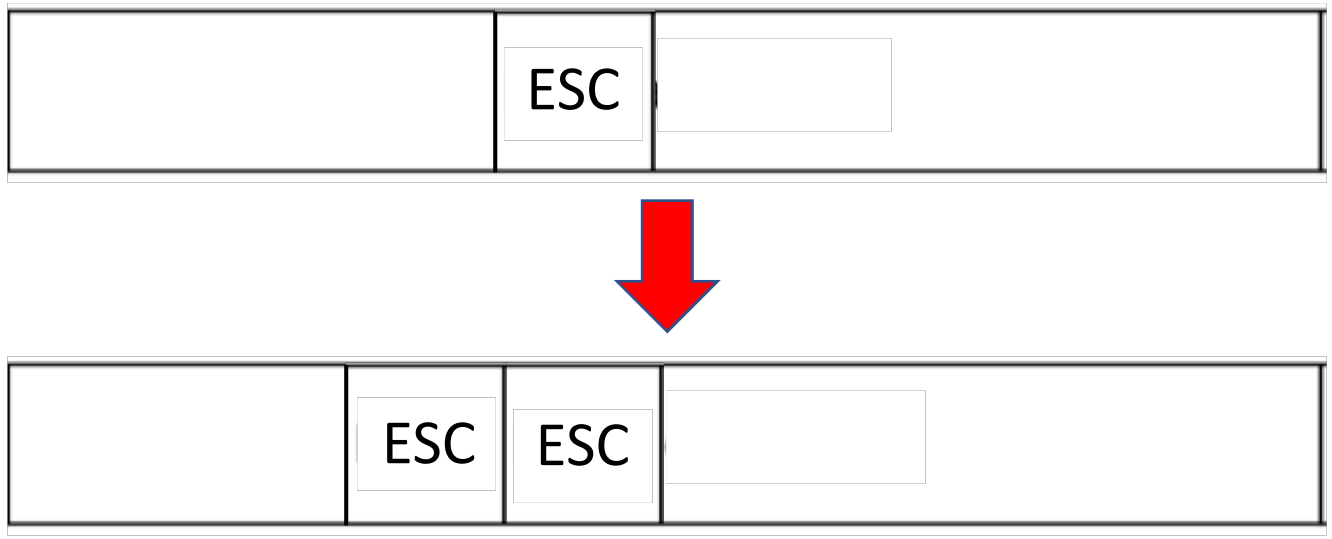

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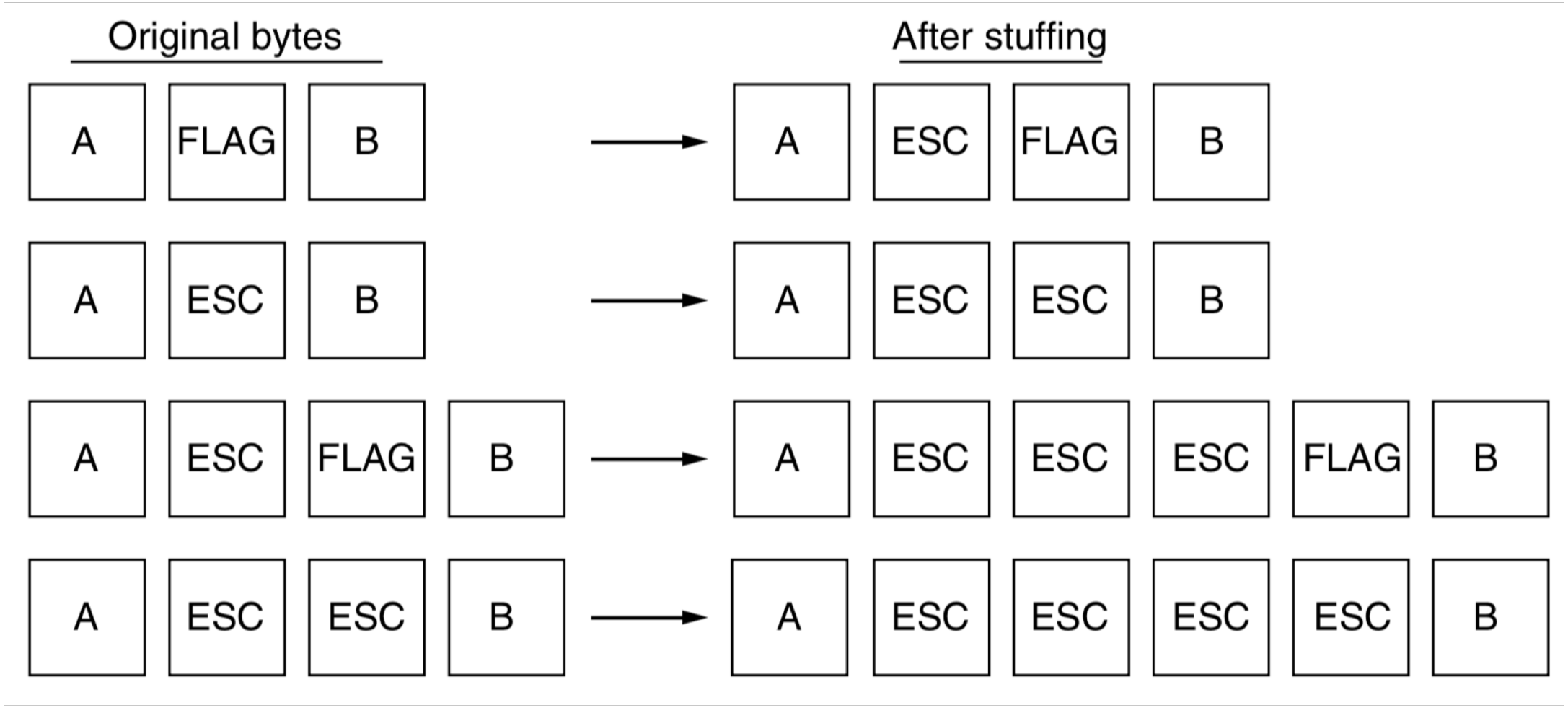
Byte Stuffing



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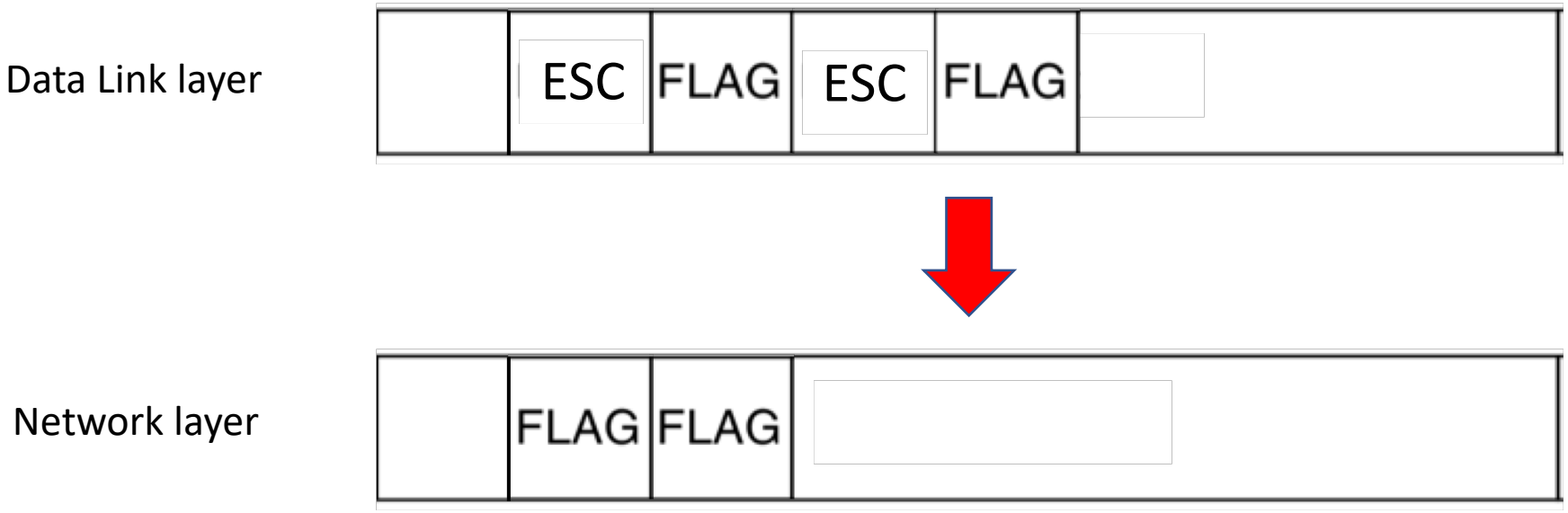
Byte Stuffing

- a framing flag byte can be distinguished from one in the data by the **absence** or **presence** of an escape byte before it. The data link layer on the **receiving end removes the escape bytes** before giving the data to the network layer.



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Byte Stuffing

- In Byte stuffing, we use a byte consisted by 8 bits to indicate FLAG or ESC existing in frame data, which cost expensive on space consumption.

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Bit Stuffing

- Each frame begins and ends with a special bit pattern, **`01111110`** or **0x7E** in hexadecimal. This pattern is a **flag byte**.
- Whenever the sender's data link layer encounters **five consecutive 1s** in the data, it automatically **stuffs a 0 bit** into the outgoing bit stream.

`01111110` → `011111010`

- This **bit stuffing** is analogous to byte stuffing, in which an escape byte is stuffed into the outgoing character stream before a flag byte in the data.

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Bit Stuffing

- Original data: `01101111111111111110010`
- Bit stuffing: `01101111101111101111101 0010`
- Receiver: `01101111111111111110010`

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Question 5

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Solution 5

1. 00000101 01000111 11100011 11100000 01111110

   5 A B ‘ESC’ ‘FLAG’

2. 01111110 01000111 11100011 11100000 11100000 11100000 01111110 01111110

   FLAG A B ESC ‘ESC’ ESC ‘FLAG’ FLAG

3. 01111110 01000111 110100011 111000000 011111010 01111110

   FLAG A B ‘ESC’ ‘FLAG’ FLAG

Question 6

- The following data fragment occurs in the middle of a data stream for which the byte-stuffing algorithm as described in the lecture is used:

A B ESC C ESC FLAG FLAG D.

- What is the output after stuffing?

Solution 6

After stuffing we get:

A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D.

Question 7

- A bit string, `0111101111101111110`, needs to be transmitted at the data link layer.

- What is the actual string transmitted across the physical transmission medium, assuming bit stuffing is used?

Solution 7

The output is `0111101111100111110 10`.